

Decomposing algebraic numbers into sums and products of minimal degree

A unified account of the theory, with complete algorithms for sums and for two-factor products, and an implementation in the Wolfram Language and in Python

Prepared for Vladimir Reshetnikov
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Abstract

Given an algebraic number a of degree n , we ask for a representation of a as a sum, or as a product, of algebraic numbers whose largest degree is as small as possible. This article unifies nine independent technical reports written on the question and adds several results. We prove that the additive problem has a complete finite algorithm (trace descent to the splitting field, then rational linear algebra on fixed fields of subgroups of the Galois group), that the two-factor multiplicative problem has a complete finite algorithm (norm descent with an explicit radical exponent), and that the arbitrary-length multiplicative problem is different: optimal factors may lie outside the splitting field, two-factor optima are not many-factor optima, and the methods developed here do not give a complete algorithm for arbitrary products. We prove rigorous lower bounds valid against any finite number of components, including a computable bound from Frobenius cycle types, and we settle the two examples of the question: both have globally optimal maximum degree three. We also answer a further question: a `Root` can be a flat sum of `Roots` of smaller degree, even with Gaussian rational coefficients, while no sequence of binary additive or multiplicative splittings into two `Roots` of smaller degree exists; the number $\sqrt{2} + \sqrt{3} + \sqrt{6}$ is such an example. The accompanying Wolfram Language package and its Python counterpart implement the principal algorithms with an engine based on numerical resolvents for the Galois group and a trace-form basis of the splitting field. This avoids the slow number-field conversions proposed in the reports; exact identity verification and the certification of exhaustive searches are discussed separately.

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1 Introduction

The question [1] asks, for a polynomial `Root` object such as

$$\begin{aligned}\alpha_{\times} &= \text{Root}[-1 - \# + 3\#^3 - \#^4 + \#^5 - 3\#^6 + 2\#^7 + \#^9 \&, 1], \\ \alpha_{+} &= \text{Root}[8 - 4\# + 24\#^2 - 15\#^3 + 3\#^5 + 6\#^6 + \#^9 \&, 1],\end{aligned}\tag{1}$$

for a representation as a product, respectively a sum, of `Root` objects of the smallest possible maximum degree. The expected answers are

$$\begin{aligned}\alpha_{\times} &= \text{Root}[1 + \# + \#^3 \&, 1] \cdot \text{Root}[1 - \# + \#^3 \&, 1], \\ \alpha_{+} &= \text{Root}[1 + \# + \#^3 \&, 1] + \text{Root}[1 - \# + \#^3 \&, 1].\end{aligned}\tag{2}$$

The constructive answer on the site [1] guesses the shape of the answer (two cubics), forms the characteristic polynomial of a Kronecker product or Kronecker sum of companion matrices with symbolic coefficients, and solves the resulting coefficient equations. This finds the decomposition once its shape is known, but it neither discovers the shape nor proves that no better one exists.

Nine technical reports ([3, 4, 5, 6, 7, 8, 9, 10, 11], reproduced verbatim in the repository) were written on the question. They agree on the core theory and differ in emphasis, examples and implementation. None of them executed its Wolfram Language code, and the number-field construction they all propose turns out to be impractical in Mathematica (Section 11). This article gives one coherent account of the mathematics, proves everything that is used, records the examples and counterexamples that the reports contributed, adds a treatment of binary versus flat decompositions and of Gaussian rational coefficients, and describes an implementation that has actually been run.

1.1 Summary of results

Write $\deg(b) = [\mathbb{Q}(b) : \mathbb{Q}]$ for the absolute degree of an algebraic number b , and define

$$D_+(a) = \min \left\{ \max_i \deg(b_i) : a = b_1 + \cdots + b_r, r \geq 1 \right\},$$

$$D_\times(a) = \min \left\{ \max_i \deg(b_i) : a = b_1 \cdots b_r, r \geq 1 \right\},$$

the components ranging over all algebraic numbers. The main results are the following.

- (i) *Lower bounds* (Section 5). Every prime divisor of $\deg(a)$ is at most $D_+(a)$ and at most $D_\times(a)$; the exponent of the Galois group G of the normal closure of $\mathbb{Q}(a)$ divides $\text{lcm}(1, \dots, d)$ for $d = D_+(a)$ or $d = D_\times(a)$; and orders of Frobenius elements, read off from factorization patterns modulo primes, give a rigorous lower bound without computing G .
- (ii) *Sums* (Section 6). $D_+(a) \leq d$ if and only if a lies in the $\mathbb{Q}$ -linear span of the subfields of degree at most d of the splitting field L of a . Equivalently, a lies in the span of the subset sums of its conjugates of degree at most d . This is a complete finite algorithm.
- (iii) *Two-factor products* (Section 8). $a = bc$ with $\deg b, \deg c \leq d$ if and only if there are subfields $E, F \subseteq L$ and an integer $t \geq 1$ with $t[E : \mathbb{Q}] \leq d$, $t[F : \mathbb{Q}] \leq d$ and $E \cap a^t F \neq \{0\}$. The exponent t cannot be dropped: for $a = \sqrt{(1 + \sqrt{2})(1 + \sqrt{3})}$ the optimal factors $\sqrt{1 + \sqrt{2}}$ and $\sqrt{1 + \sqrt{3}}$ lie outside L , and no elements of L of degree below 8 multiply or add up to a .
- (iv) *Many factors* (Section 9). $D_\times$ can be smaller than the two-factor minimum: $(1 + \sqrt{2})(1 + \sqrt{3})(1 + \sqrt{5})$ has $D_\times = 2$ but two-factor minimum 4. Products inside a fixed field are a multiplicative, not a linear, problem; arithmetic obstructions exist beyond Galois theory ($D_+(1 + \sqrt{2} + \sqrt{3}) = 2$ but $D_\times(1 + \sqrt{2} + \sqrt{3}) = 4$). We give the necessary norm condition, a sufficient radical condition, and an exact rank-one tensor criterion for linearly disjoint families; a complete many-factor decision procedure remains outside this work.
- (v) *Binary versus flat* (Section 7). A decomposition into two components of smaller degree, iterated, is strictly weaker than a flat decomposition: $\sqrt{2} + \sqrt{3} + \sqrt{6}$ is a sum of three quadratic numbers but is neither a sum nor a product of two algebraic numbers of degree below four, even if Gaussian rational coefficients are allowed at every binary step. With Gaussian coefficients the flat additive problem is again linear algebra, in $L(i)$.
- (vi) *The examples* (Section 4). Both numbers in (1) have globally optimal maximum degree 3, attained by (2). Moreover $D_+(\alpha_\times) = 6$, attained by two sextics that do not lie in $\mathbb{Q}(\alpha_\times)$, whereas every additive decomposition of $\alpha_\times$ inside $\mathbb{Q}(\alpha_\times)$ has maximum degree 9.
- (vii) *Implementation* (Section 11). The Galois group is computed by numerical resolvents, the splitting field is represented in a tower basis with coordinates obtained from traces, and the additive and two-factor criteria become rational linear algebra once the field data are certified. The many-factor search also needs multiplicative tests. A fast path works inside $\mathbb{Q}(a)$ without the Galois group. The original measurements answered the two examples in about a second in Wolfram (fast path), certifying global optimality; the Python port (python-flint, Arb ball arithmetic) constructed the full splitting-field data in three to five seconds, against about twenty seconds in the Wolfram kernel. Section 11 distinguishes these historical timings from validation of the revised code.

1.2 Conventions

All fields are subfields of $\mathbb{C}$, so every algebraic extension is separable. A number field is a finite extension of $\mathbb{Q}$. For a finite Galois extension $L/\mathbb{Q}$ with group $G = \text{Gal}(L/\mathbb{Q})$, subfields of L correspond to subgroups $H \leq G$ via $H \mapsto L^H$, and $[L^H : \mathbb{Q}] = [G : H]$. The degree of an element $b \in L$ is the index of its stabilizer, equivalently the size of its G -orbit. We write $\text{Res}_y(f, g)$ for the resultant with respect to y , and $\text{Tr}_{M/L}, \text{N}_{M/L}$ for relative trace and norm.

Mathematica's `Root[f, k]` indexes real roots first, in increasing order, and places non-real conjugate pairs consecutively for rational polynomials. Its documentation warns that the order of non-real roots can depend on the root-isolation method [2]. The Python code uses its own explicit order for those pairs: increasing real part, then increasing absolute imaginary part, with the negative imaginary part first. Thus real-root indices agree, including the two original examples, but an arbitrary non-real branch must be matched by an isolating region when moving between implementations; a printed index alone is not a universal interchange format.

2 The problem

Definition 2.1. For $a \in \overline{\mathbb{Q}}$ put

$$D_+(a) = \min\{\max_i \deg(b_i) : r \geq 1, b_i \in \overline{\mathbb{Q}}, a = b_1 + \cdots + b_r\},$$

$$D_\times(a) = \min\{\max_i \deg(b_i) : r \geq 1, b_i \in \overline{\mathbb{Q}}^\times, a = b_1 \cdots b_r\} \quad (a \neq 0),$$

and $D_\times(0) = 1$. Let $D_{+,2}(a)$ and $D_{\times,2}(a)$ be the same minima restricted to $r \leq 2$. For a field $K \ni a$ let $D_+^K(a), D_\times^K(a)$ be the minima with all components in K .

The one-component representation shows that all these quantities are at most $\deg(a)$, and $D_\times \leq D_{\times,2}, D_+ \leq D_{+,2}, D_+ \leq D_+^K, D_\times \leq D_\times^K$. Several elementary remarks fix the conventions.

- (a) Degrees are absolute. A displayed `Root` may be a root of a reducible polynomial, and a polynomial of small degree with algebraic coefficients says nothing about absolute degree: $x - a$ has degree one over $\mathbb{Q}(a)$. Degrees are therefore measured with the rational minimal polynomial.
- (b) Rational scalars are free: for $q \in \mathbb{Q}^\times, \mathbb{Q}(qb) = \mathbb{Q}(b)$. A rational linear combination of low-degree numbers is a permitted sum, and a rational multiple of a factor may be moved to another factor. Rational numbers have degree one. Inverses and negatives do not change degrees.
- (c) Components are allowed to be complex even when a is real; requiring real components is a different problem (Remark 6.7).
- (d) The number of components, the coefficient heights and the printed size of the answer are not minimized; they are secondary objectives. A flat sum $b_1 + \cdots + b_r$ is not a nested expression such as $b_1 + b_2 b_3$; Section 7 compares the two.
- (e) Three ambient domains matter: $K = \mathbb{Q}(a) \subseteq L \subseteq \overline{\mathbb{Q}}$, where L is the splitting field of the minimal polynomial of a . For sums, L is always large enough; for products not even L is (Section 8). Restricting to K can change both answers.

3 Composed sums and products

Let $f, g \in \mathbb{Q}[x]$ be monic of degrees m, k with roots $u_1, \dots, u_m$ and $v_1, \dots, v_k$ (with multiplicity). Their *composed sum* and *composed product* are

$$\begin{aligned} (f \oplus g)(x) &= \text{Res}_y(f(y), g(x-y)) = \prod_{i,j} (x - u_i - v_j), \\ (f \otimes g)(x) &= \text{Res}_y(f(y), y^k g(x/y)) = \prod_{i,j} (x - u_i v_j). \end{aligned} \tag{3}$$

Both identities follow from $\text{Res}_y(f, h) = \prod_i h(u_i)$ for monic f ; the second argument of the product resultant is the polynomial $\sum_j g_j x^j y^{k-j}$, which handles a zero root correctly. Equivalently, $f \oplus g$ and $f \otimes g$ are the characteristic polynomials of $C_f \otimes I + I \otimes C_g$ and $C_f \otimes C_g$, where C_f, C_g are companion matrices; this is the construction used in the answer on the site. Fast algorithms for composed sums and products exist [14]; here only small degrees occur.

Proposition 3.1. *Let p be the minimal polynomial of a . Some pair of roots (u_i, v_j) satisfies $a = u_i + v_j$ (respectively $a = u_i v_j$) if and only if p divides $f \oplus g$ (respectively $f \otimes g$).*

Proof. If $a = u_i + v_j$ then a is a root of $f \oplus g$, and the irreducible p divides it. Conversely, if $p \mid f \oplus g$ then a is a root of the product in (3), so some linear factor vanishes at a . The product case is identical. $\square$

Divisibility cannot be replaced by equality: $(x^2 - 2) \oplus (x^2 - 2) = x^2(x^2 - 8)$ and $(x^2 - 2) \otimes (x^2 - 3) = (x^2 - 6)^2$, while $\sqrt{2} + \sqrt{2}$ and $\sqrt{2} \cdot \sqrt{3}$ have minimal polynomials $x^2 - 8$ and $x^2 - 6$. Nor need $\deg(a)$ divide mk : with $u = \sqrt[3]{2}$ and $\omega = e^{2\pi i/3}$ the number $u - \omega u$ has minimal polynomial $x^6 + 108$, of degree $6 \nmid 9$. The correct general statement is $\deg(a) \leq [\mathbb{Q}(b_1, \dots, b_r) : \mathbb{Q}] \leq \prod \deg(b_i)$.

For the two cubics of (2) one computes

$$\begin{aligned} (x^3 + x + 1) \oplus (x^3 - x + 1) &= x^9 + 6x^6 + 3x^5 - 15x^3 + 24x^2 - 4x + 8, \\ (x^3 + x + 1) \otimes (x^3 - x + 1) &= x^9 + 2x^7 - 3x^6 + x^5 - x^4 + 3x^3 - x - 1, \end{aligned} \tag{4}$$

exactly the polynomials in (1).

3.1 Inverse composition: fix one polynomial

Guessing both polynomials with unknown coefficients leads to a nonlinear system whose solutions over $\mathbb{C}$ may have algebraic coefficients (which do not certify absolute degrees) and whose normalizations need justification (rescaling $u \mapsto qu$, $v \mapsto v/q$ with $q \in \mathbb{Q}^\times$ preserves a product, but making a constant term equal to 1 requires a rational m th root). A cheaper exact device fixes one candidate polynomial f and recovers the partner: if $a = u + v$ with $f(u) = 0$ then v is a root of $\text{Res}_y(f(y), p(x+y))$, and if $a = uv$ then v is a root of $\text{Res}_y(f(y), p(xy))$ (for $f(0) \neq 0$). Factoring these resultants over $\mathbb{Q}$ and keeping the factors of degree at most d gives a complete search for a fixed first polynomial. For $f = x^3 + x + 1$ and both examples the resultant factors as $(x^3 - x + 1)^3 h_{18}(x)$ with h_{18} irreducible [4, 6]. Enumerating first polynomials of bounded degree and coefficient height, including non-monic primitive polynomials (otherwise $\frac{1}{2}$ would be missed), gives the *bounded dictionary search*: complete within its box, a positive semi-decision procedure in general, never a proof of impossibility. Even the smallest box, cubics with coefficients in $\{-1, 0, 1\}$, discovers both cubics of (2).

4 The examples: identities, degrees, branches

Let u and v be the real roots of $f = x^3 + x + 1$ and $g = x^3 - x + 1$. Both cubics are irreducible (no rational roots) with discriminants -31 and -23 , so each has exactly one real root and Galois group S_3 . Write $P_\times$ and P_+ for the two polynomials in (1).

Proposition 4.1. $P_\times$ and P_+ are irreducible, each has exactly one real root, and $\alpha_\times = uv$, $\alpha_+ = u + v$. Moreover $\mathbb{Q}(uv) = \mathbb{Q}(u + v) = \mathbb{Q}(u, v)$ has degree nine, and

$$u = \frac{\alpha_\times^3 - \alpha_\times^2 - 1}{\alpha_\times^4 + \alpha_\times^2 - \alpha_\times + 1}, \quad v = \frac{\alpha_\times}{u}; \quad u = \frac{3\alpha_+^5 - 5\alpha_+^3 - 3\alpha_+^2 + 2\alpha_+ - 4}{6\alpha_+^4 - 6\alpha_+ + 4}, \quad v = \alpha_+ - u. \quad (5)$$

Also $u = -(\alpha_\times^7 + 2\alpha_\times^5 - 2\alpha_\times^4 + 1)/2$ and $v = -(\alpha_\times^7 + 2\alpha_\times^5 - 2\alpha_\times^4 + 2\alpha_\times^3 + 2\alpha_\times - 1)/2$.

Proof. The splitting fields L_f, L_g of the cubics are S_3 -extensions whose unique quadratic subfields $\mathbb{Q}(\sqrt{-31})$ and $\mathbb{Q}(\sqrt{-23})$ differ. A nontrivial common Galois quotient of two copies of S_3 is C_2 or S_3 , and either would force a common quadratic subfield; hence $L_f \cap L_g = \mathbb{Q}$ and $\text{Gal}(L_f L_g / \mathbb{Q}) = S_3 \times S_3$, acting transitively on the nine ordered pairs (u_i, v_j) of roots. If two pairs had the same sum, $u_i - u_k = v_l - v_j \in L_f \cap L_g = \mathbb{Q}$; a nonzero rational difference of two conjugates is impossible, since iterating an automorphism mapping $u_k \mapsto u_i$ would produce arbitrarily large differences. If two pairs had the same product, $u_i/u_k \in \mathbb{Q}^\times$ would be a root of unity, hence ± 1 , and -1 is excluded because $f(x) + f(-x) = 2 \neq 0$. So the nine sums and the nine products are distinct, the resultants (4) are the minimal polynomials, and both are irreducible of degree nine. A real conjugate of uv or $u + v$ is fixed by complex conjugation, which permutes the pairs; the only pair fixed by conjugation is (u, v) , so each polynomial has exactly one real root, namely uv , respectively $u + v$; by the root ordering this is $\text{Root}[\dots, 1]$.

The recovery formulas are verified by exact polynomial arithmetic modulo $P_\times$ and P_+ : the denominators are coprime to the respective polynomial, and substituting the right-hand sides into f and g gives zero modulo $P_\times$, respectively P_+ . The real-branch identification follows from the uniqueness of the real roots of the two cubics. These checks show $u, v \in \mathbb{Q}(\alpha_\times)$ and $u, v \in \mathbb{Q}(\alpha_+)$; since $[\mathbb{Q}(u, v) : \mathbb{Q}] = 9$, all three fields coincide. $\square$

5 Lower bounds valid for any number of components

Theorem 5.1 (normal closure obstruction). *Let $n = \deg(a)$, let L be the splitting field of the minimal polynomial of a and $G = \text{Gal}(L/\mathbb{Q})$. Suppose a lies in the field generated by finitely many algebraic numbers $b_1, \dots, b_r$ of degrees at most d (in particular, if a is a sum or a product of them). Then*

- (i) every prime divisor of $|G|$, hence of n , is at most d ;
- (ii) $\exp(G)$ divides $\text{lcm}(1, 2, \dots, d)$; in particular $d \geq$ the largest prime power dividing $\exp(G)$;
- (iii) if $d \leq 4$ then G is solvable.

Consequently $D_+(a)$ and $D_\times(a)$ are at least the largest prime divisor of n and at least the largest prime power dividing $\exp(G)$.

Proof. Let M_i be the normal closure of $\mathbb{Q}(b_i)$ and $M = M_1 \cdots M_r$. $\text{Gal}(M_i/\mathbb{Q})$ acts faithfully on the $m_i \leq d$ conjugates of b_i , so it embeds in S_{m_i} , and restriction embeds $\text{Gal}(M/\mathbb{Q})$ into $\prod_i \text{Gal}(M_i/\mathbb{Q}) \leq \prod_i S_{m_i}$. Every prime divisor of $|\prod_i S_{m_i}| = \prod_i m_i!$ is at most d , the exponent of $\prod_i S_{m_i}$ divides $\text{lcm}(1, \dots, d)$ (orders of permutations are lcm's of cycle lengths), and S_m is solvable for $m \leq 4$. These properties pass to subgroups and to quotients. Since $M/\mathbb{Q}$ is normal and contains a , it contains all conjugates of a , hence L , and restriction maps $\text{Gal}(M/\mathbb{Q})$ onto G . Finally $n = [\mathbb{Q}(a) : \mathbb{Q}]$ divides $|G| = [L : \mathbb{Q}]$. $\square$

For the examples, $n = 9$ gives the bound 3, which (2) attains; so $D_\times(\alpha_\times) = D_+(\alpha_+) = 3$ globally, against any finite number of complex components. The exponent bound is genuinely stronger than the prime bound: for ζ_5 ($G = C_4$) and for $\sqrt{2 + \sqrt{2}}$ ($G = C_4$) it gives $D_+ = D_\times = 4$ although the prime bound is 2; for $2^{1/4}$ ($G = D_4$, exponent 4) it gives 4; for a root of $x^4 - x - 1$ ($G = S_4$, exponent 12) it gives 4.

Proposition 5.2 (Frobenius bound). *Let $P \in \mathbb{Z}[x]$ be the primitive minimal polynomial of a and let q be a prime dividing neither the leading coefficient nor the discriminant of P . If P factors modulo q into irreducible factors of degrees $e_1, \dots, e_s$, then G contains an element of order $\text{lcm}(e_1, \dots, e_s)$, so this number divides $\exp(G)$. Hence $D_+(a), D_\times(a) \geq$ the largest prime power dividing $\text{lcm}_q \text{lcm}(e_1, \dots, e_s)$, the outer lcm over any finite set of such primes.*

Proof. For unramified q the Frobenius element at a prime of L above q acts on the roots with cycle type $(e_1, \dots, e_s)$ (Dedekind's theorem, e.g. [15, Th. 4.37]), hence has order $\text{lcm}(e_1, \dots, e_s)$. $\square$

The Frobenius bound is cheap (a few factorizations modulo small primes) and is the bound the implementations use before any Galois group is computed. For $x^4 - x - 1$ the factorizations modulo 2 and 7 have patterns (4) and (1, 3), giving exponent multiple 12 and the bound 4. In general the bound may be weaker than the true exponent bound; the full Galois data, when available, sharpens it.

Proposition 5.3 (length-dependent bound). *If $a = b_1 \circ \dots \circ b_r$ (sums or products) with $\deg(b_i) \leq d$, then $n \leq d^r$. In particular a two-component representation needs $d \geq \lceil \sqrt{n} \rceil$.*

Proof. $\mathbb{Q}(a) \subseteq \mathbb{Q}(b_1, \dots, b_r)$, whose degree is at most $\prod \deg(b_i)$. $\square$

The bounds of this section are necessary conditions. They are not sufficient even for sums: $\alpha_\times$ has $D_+ = 6$ (Example 6.5), while its degree-nine prime bound and its $S_3 \times S_3$ exponent bound are both 3. Nor are the bounds sufficient for products even when the Galois group is elementary abelian (Theorem 9.7).

6 Sums: a complete finite algorithm

6.1 Trace descent

Lemma 6.1 (trace projection). *Let $L/\mathbb{Q}$ be a finite Galois extension, $a \in L$, and $a = b_1 + \dots + b_r$ with arbitrary algebraic b_i . Then there are $c_i \in L \cap \mathbb{Q}(b_i)$ with $a = c_1 + \dots + c_r$; in particular $\deg(c_i) \leq \deg(b_i)$.*

Proof. Choose a finite Galois extension $M/\mathbb{Q}$ containing L and all b_i ; put $\Gamma = \text{Gal}(M/\mathbb{Q})$ and $N = \text{Gal}(M/L)$, a normal subgroup of Γ since $L/\mathbb{Q}$ is Galois. Define the $\mathbb{Q}$ -linear projection

$$\pi(z) = \frac{1}{|N|} \sum_{\sigma \in N} \sigma(z) = \frac{\text{Tr}_{M/L}(z)}{[M:L]}, \quad M \rightarrow L,$$

which fixes L pointwise. Normality of N gives $\gamma\pi(z) = \pi(\gamma z)$ for all $\gamma \in \Gamma$: conjugation by γ permutes N . Hence every automorphism fixing b_i fixes $\pi(b_i)$, and by the Galois correspondence $\pi(b_i) \in \mathbb{Q}(b_i)$. Put $c_i = \pi(b_i) \in L \cap \mathbb{Q}(b_i)$; then $\sum c_i = \pi(a) = a$. $\square$

Normality of L is essential: averaging over $\text{Gal}(M/\mathbb{Q}(a))$ for non-normal $\mathbb{Q}(a)$ is not Γ -equivariant, and Example 6.6 shows that restricting summands to $\mathbb{Q}(a)$ can change the answer.

6.2 The fixed-space criterion

For L Galois with group G and $d \geq 1$ define the $\mathbb{Q}$ -vector subspace

$$W_d(L) = \sum_{H \leq G, [G:H] \leq d} L^H \subseteq L, \tag{6}$$

a sum of subspaces, not a compositum of fields.

Theorem 6.2 (complete additive criterion). *Let $a \in L$ with $L/\mathbb{Q}$ finite Galois. Then $D_+(a) \leq d$ if and only if $a \in W_d(L)$. In particular $D_+(a) = \min\{d : a \in W_d(L)\}$, and this is computable by finitely many rational linear-algebra operations once G and its subgroups are known. If a is not rational the minimum is attained by a representation with at most $[L : \mathbb{Q}]$ nonzero summands.*

Proof. If $a = \sum b_i$ with $\deg(b_i) \leq d$, Lemma 6.1 gives $a = \sum c_i$ with $c_i \in L$ and $\deg(c_i) \leq d$; each c_i lies in the subfield $\mathbb{Q}(c_i) = L^{H_i}$ with $[G : H_i] \leq d$, so $a \in W_d(L)$. Conversely an element of $W_d(L)$ is a finite sum of elements of fields of degree at most d , each of which has degree at most d . Choosing a basis of $W_d(L)$ from the union of bases of the fields and absorbing the rational coefficients gives at most $\dim W_d(L) \leq [L : \mathbb{Q}]$ summands. The minimum is attained at some $d \leq \deg(a)$ because $\mathbb{Q}(a) \subseteq L$ is itself one of the fields. $\square$

Only the subgroups H that are minimal among those of index at most d matter, since $L^{H'} \supseteq L^H$ for $H' \leq H$. Writing B_d for the matrix whose columns are the coordinate vectors of bases of these fields, $a \in W_d(L)$ if and only if $B_d c = \text{vec}(a)$ has a rational solution, and the solution grouped by field gives the summands.

A failed membership test also has a small exact certificate: a rational row vector λ with $\lambda B_d = 0$ but $\lambda \text{vec}(a) \neq 0$. Such a separating functional exists precisely when the system is inconsistent. Checking it only requires rational matrix multiplication, although the claim that it rules out every allowed summand still requires the field list to be complete.

6.3 The permutation-module reformulation

The criterion can be pushed down from L to the $\mathbb{Q}$ -span of the conjugates of a , which has dimension at most n and needs no basis of L . Let $a_1, \dots, a_n$ be the conjugates of a and $V = \text{span}_{\mathbb{Q}}\{a_1, \dots, a_n\} \subseteq L$, a G -stable subspace. For $S \subseteq \{1, \dots, n\}$ write $s_S = \sum_{i \in S} a_i$.

Theorem 6.3 (conjugate-subset criterion, [3]). *$D_+(a) \leq d$ if and only if a lies in the $\mathbb{Q}$ -span of the subset sums s_S with $\deg(s_S) \leq d$.*

Proof. Since $\mathbb{Q}[G]$ is semisimple, V has a G -stable complement in L and there is a G -equivariant projection $P : L \rightarrow V$ (average any projection over G). P is the identity on V , so $P(a) = a$, and $P(L^H) \subseteq V^H$ for every H . Hence $a \in W_d(L)$ implies $a \in \sum_{[G:H] \leq d} V^H$; the converse holds because $V^H \subseteq L^H$. Finally V^H is spanned by the H -orbit sums of the a_i : the surjection $\mathbb{Q}^n \rightarrow V$, $e_i \mapsto a_i$, is G -equivariant, so by averaging lifts over H it restricts to a surjection $(\mathbb{Q}^n)^H \rightarrow V^H$, and $(\mathbb{Q}^n)^H$ is spanned by the indicator vectors of the H -orbits. Each orbit sum is fixed by H and so has degree at most $[G : H] \leq d$. $\square$

Thus the additive problem lives entirely in the permutation module of G on the roots. The only information beyond the group that it needs is the space of $\mathbb{Q}$ -linear relations among the conjugates (the kernel of $\mathbb{Q}^n \rightarrow V$), which the implementation obtains from the trace form (Section 11).

6.4 Examples

Example 6.4 (the sum example). For α_+ the splitting field is $L = L_f L_g$ of degree 36 with $G = S_3 \times S_3$. The subgroup counts by index are 1, 3, 6, 1, 20, 9, 4, 15, 1 for indices 1, 2, 3, 4, 6, 9, 12, 18, 36 (sixty subgroups). The six cubic subfields are the fields generated by the conjugates of u and of v , and $\alpha_+ \in \mathbb{Q}(u) + \mathbb{Q}(v)$ recovers (2). Since 3 is also the prime lower bound, the answer is globally optimal.

Example 6.5 (the product root as a sum). The same field serves $\alpha_{\times} = uv$. Inside $K = \mathbb{Q}(uv)$, whose subfields have degrees 1, 3, 3, 9 (they are $\mathbb{Q}$, $\mathbb{Q}(u)$, $\mathbb{Q}(v)$, K), the sum $\mathbb{Q}(u) + \mathbb{Q}(v) = \text{span}\{1, u, u^2, v, v^2\}$ does not contain uv because $\{u^i v^j\}_{0 \leq i, j \leq 2}$ is a basis of K ; so $D_+^K(uv) = 9$.

Globally, label the roots $u_1 = u, u_2, u_3$ and $v_1 = v, v_2, v_3$ and put $w_\pi = \sum_j u_j v_{\pi(j)}$ for $\pi \in S_3$. The two permutations fixing 1 give real numbers w_1, w_2 with

$$w_1 + w_2 = 2uv + (u_2 + u_3)(v_2 + v_3) = 3uv$$

because both cubics have zero trace. The six values w_π are the roots of the irreducible sextic

$$R(x) = x^6 + 6x^4 - 27x^3 + 9x^2 - 81x + 4 = \left(x^3 + 3x - \frac{27}{2}\right)^2 - \frac{713}{4}, \quad (7)$$

which has exactly two real roots; indeed w_1, w_2 satisfy $w^2 - 3uvw + 3u^2 - 3v^2 + 4 = 0$ and eliminating u, v with the two cubics gives $R(w)^3$. Hence

$$\begin{aligned} \alpha_\times &= \frac{1}{3}\text{Root}[R, 1] + \frac{1}{3}\text{Root}[R, 2] \\ &= \text{Root}[S, 1] + \text{Root}[S, 2], \quad S(x) = R(3x), \end{aligned}$$

a sum of two sextics, and $D_+(uv) \leq 6$. To see that 6 is optimal, let $A = C_3 \times C_3 \leq G$ be the Sylow 3-subgroup, $U = \text{span}\{u_1, u_2, u_3\}$ and $V = \text{span}\{v_1, v_2, v_3\}$; these are the two-dimensional standard representations of the two factors, and linear disjointness of L_f, L_g makes multiplication $U \otimes V \rightarrow L$ injective, so $T = \text{span}\{u_i v_j\}$ is a four-dimensional irreducible G -module containing uv . A subgroup H of index at most 5 has index 1, 2, 3 or 4. If the index is 1, 2 or 4 then $H \supseteq A$ (the index-4 subgroup is A itself), and A has no nonzero fixed vector in T . If the index is 3, then $|H| = 12$, $H \cap A$ has order 3, and H maps onto $G/A = C_2 \times C_2$; the two independent sign changes act on $A \cong \mathbb{F}_3^2$ by inverting one coordinate each, and a line of $\mathbb{F}_3^2$ invariant under both is a coordinate axis, so $H \supseteq C_3 \times 1$ or $1 \times C_3$, again giving $T^H = 0$. The equivariant projection $L \rightarrow T$ kills every L^H with $[G : H] \leq 5$ and fixes uv , so $uv \notin W_5(L)$. Therefore $D_+(\alpha_\times) = 6$, while $D_+^K(\alpha_\times) = 9$ and $D_\times(\alpha_\times) = 3$.

Example 6.6 (a sextic needing quartic summands outside its own field). Let $r_1 < r_2$ be the real roots of $x^4 - x - 1$ and $s = r_1 + r_2$. Then $\text{Res}_y(y^4 - y - 1, (x - y)^4 - (x - y) - 1) = (x^4 - 8x - 16)(x^6 + 4x^2 - 1)^2$ and s is the positive root of the irreducible sextic $x^6 + 4x^2 - 1$. The quartic is irreducible modulo 2 and has pattern (1, 3) modulo 7, so its Galois group contains a 4-cycle and a 3-cycle and is S_4 ; the stabilizer of the pair $\{r_1, r_2\}$ is $\langle (12), (34) \rangle$ of index 6, whose only proper overgroup is the dihedral group of order 8. So $\mathbb{Q}(s)$ has subfields of degrees 1, 3, 6 only and $D_+^{\mathbb{Q}(s)}(s) = 6$; but $s = r_1 + r_2$ shows $D_+(s) \leq 4$, and Theorem 5.1(ii) with $\exp(S_4) = 12 \nmid \text{lcm}(1, 2, 3)$ gives $D_+(s) = 4$. For the stated real branches the identity is

$$\text{Root}[-1 + 4\#^2 + \#^6 \&, 2] = \text{Root}[-1 - \# + \#^4 \&, 1] + \text{Root}[-1 - \# + \#^4 \&, 2].$$

Remark 6.7 (real summands). If a is real and one insists on real summands, the criterion is obtained by restricting (6) to subgroups containing complex conjugation: trace projection commutes with conjugation, so it preserves reality.

7 Binary splittings, flat sums, and Gaussian rational coefficients

A natural strategy for producing decompositions is to split a into *two* Roots of smaller degree and to iterate on the pieces. The following question was raised while this article was written: can a Root be a linear combination of Roots of smaller degree with Gaussian rational coefficients, while no such iterated binary splitting exists? The answer is yes, and the phenomenon already occurs with rational coefficients.

Definition 7.1. A *binary additive step* for a is a representation $a = b + c$ with $\deg b, \deg c < \deg a$; a *binary multiplicative step* is $a = bc$ with the same condition. A *binary tree* for a applies such steps recursively. Its leaves are components of degree smaller than $\deg a$; the first step alone already gives a decomposition into two components of smaller degree.

Proposition 7.2 (binary criteria). *Let L be the splitting field of a , $n = \deg a$. A binary additive step exists if and only if $a \in E_1 + E_2$ for two subfields $E_1, E_2 \subseteq L$ of degree less than n . A binary multiplicative step exists if and only if the two-factor criterion of Theorem 8.2 holds with $d = n - 1$.*

Proof. For sums, apply Lemma 6.1 to $a = b + c$: the projections $\pi(b), \pi(c)$ lie in L , have degrees at most $\deg b, \deg c < n$, and add up to a . The converse is clear. The product statement is Theorem 8.2. $\square$

Theorem 7.3. *Let $\eta = \sqrt{2} + \sqrt{3} + \sqrt{6}$, of degree 4 with minimal polynomial $x^4 - 22x^2 - 48x - 23$. Then $D_+(\eta) = 2$, attained by the flat sum $\sqrt{2} + \sqrt{3} + \sqrt{6}$, but η admits no binary additive step and no binary multiplicative step. The same holds if Gaussian rational coefficients are allowed in the binary steps, that is, there are no $c, c' \in \mathbb{Q}(i)$ and algebraic X, Y of degree at most 3 with $\eta = cX + c'Y$ or $\eta = cXY$.*

Proof. $\mathbb{Q}(\eta) = L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ is Galois with group $C_2 \times C_2$; its proper subfields are $\mathbb{Q}$ and the three quadratic fields $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{6})$. In the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ the vector-space sum of any two of these fields misses at least one of the three coordinates in which η is nonzero, so $\eta \notin E_1 + E_2$ and there is no binary additive step by Proposition 7.2. For products apply Theorem 8.2 with $d = 3$: the exponent satisfies $t \leq \min(3, \lfloor 9/4 \rfloor) = 2$, and $t = 2$ forces $E = F = \mathbb{Q}$, i.e. $\eta^2 \in \mathbb{Q}$, which is false. For $t = 1$ the fields E, F are quadratic (or $\mathbb{Q}$), and one checks the three pairs directly: a product $(x_0 + x_1\sqrt{2})(y_0 + y_1\sqrt{3})$ has coordinate vector $(x_0y_0, x_1y_0, x_0y_1, x_1y_1)$, whose first coordinate vanishes only if the second or third does; the pairs $(\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{6}))$ and $(\mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{6}))$ are the same computation. So $D_{\times,2}(\eta) = 4$.

Now allow Gaussian coefficients. For sums, suppose $\eta = cX + c'Y$ with $c, c' \in \mathbb{Q}(i)$ and $\deg X, \deg Y \leq 3$. Work in the Galois field $L' = L(i) = \mathbb{Q}(i, \sqrt{2}, \sqrt{3})$, of degree 8. The projection $\pi : M \rightarrow L'$ of Lemma 6.1 is L' -linear, so $\eta = c\pi(X) + c'\pi(Y)$ with $\pi(X), \pi(Y) \in L'$ of degree at most 3, hence lying in $\mathbb{Q}$ or in a quadratic subfield $\mathbb{Q}(\sqrt{m})$, $m \in \{-1, \pm 2, \pm 3, \pm 6\}$. Take real parts: if $m > 0$ then $\operatorname{Re}(cX') = \operatorname{Re}(c)X' \in \mathbb{Q}(\sqrt{m})$, and if $m < 0$, $X' = p + qi\sqrt{|m|}$ with $p, q \in \mathbb{Q}$ and $c = e + e'i$ give $\operatorname{Re}(cX') = ep - e'q\sqrt{|m|} \in \mathbb{Q}(\sqrt{|m|})$. So η is a sum of two elements of real quadratic subfields of L , which was excluded above. For products, $\eta = cXY$ with $c \notin \mathbb{Q}$ means that η/c is a product of two numbers of degree at most 3; writing $c^{-1} = e + e'i$ with $e' \neq 0$, the element $\eta/c = e\eta + e'i\eta$ has nonzero coordinates on $i\sqrt{2}, i\sqrt{3}, i\sqrt{6}$. An automorphism sending $(i, \sqrt{2}, \sqrt{3})$ to $(\epsilon_0 i, \epsilon_1 \sqrt{2}, \epsilon_2 \sqrt{3})$ can fix it only if $\epsilon_0 \epsilon_1 = \epsilon_0 \epsilon_2 = \epsilon_0 \epsilon_1 \epsilon_2 = 1$, which forces every $\epsilon_j = 1$. Thus its stabilizer in $\operatorname{Gal}(L'/\mathbb{Q})$ is trivial and it has degree 8. Theorem 8.2 with $n = 8, d = 3$ allows only $t = 1$ and quadratic E, F , whose compositum has degree at most $4 < 8$. If $c \in \mathbb{Q}$ the rational case applies. $\square$

The same argument shows that $i + \sqrt{2} + i\sqrt{2}$, of degree 4, is the two-term Gaussian combination $i \cdot 1 + (1 + i)\sqrt{2}$ of Roots of degree 1 and 2 but admits no binary step with rational coefficients. The general principle is that a binary step tests membership in the sum of *two* subfields, whereas Theorem 6.2 allows the sum of all of them; a flat sum of k terms can require k distinct subfields none of whose pairwise sums contains a .

Theorem 7.4 (Gaussian coefficients, flat sums). *Let L' be a finite Galois extension containing a and i , with group G' . Then $a = \sum_j c_j b_j$ with $c_j \in \mathbb{Q}(i)$ and $\deg(b_j) \leq d$ for some finite r if and only if*

$$a \in \sum_{H \leq G', [G':H] \leq d} (L'^H + i L'^H).$$

Proof. The projection $\pi : M \rightarrow L'$ is L' -linear, hence $\mathbb{Q}(i)$ -linear, so $a = \sum c_j \pi(b_j)$ with $\pi(b_j)$ in subfields E_j of L' of degree at most d , and $c_j E_j \subseteq E_j + i E_j$. Conversely $E + iE$ consists of the two-term combinations $1 \cdot e + i \cdot e'$. $\square$

The Gaussian flat problem is therefore again exact linear algebra (the package option "Coefficients" -> "GaussianRationals" adjoins i to the splitting field). The lower bounds change: adjoining Gaussian coefficients adds a factor C_2 to the product of normal-closure groups in Theorem 5.1. Consequently only prime divisors ≥ 3 of n give the prime bound, while the sharper exponent condition is

$$\exp(G') \mid \text{lcm}(2, \text{lcm}(1, \dots, d)),$$

when L' is the splitting field of a with i adjoined. Indeed this field is contained in the compositum of $\mathbb{Q}(i)$ with the normal closures of the components. For $d \geq 2$ the exponent bound is unchanged; Gaussian coefficients only waive its obstruction at $d = 1$. The binary Gaussian problem $a = cX + c'Y$ is, by contrast, not linear (cE is a subspace for each fixed c , but their union over c is not); the real-part argument above settles it for real targets in totally real multiquadratic fields, and the package answers the rational binary question through the option "MaxTerms" -> 2.

8 Products of two factors: a complete algorithm

Trace is additive; norm is multiplicative but takes a to a power a^t . Keeping that power is what makes the following criterion complete.

Lemma 8.1 (intersection lemma). *Let $L/\mathbb{Q}$ be finite Galois and b algebraic. Put $E_0 = L \cap \mathbb{Q}(b)$ and $t = [L(b) : L]$. Then $\deg(b) = t[E_0 : \mathbb{Q}]$, the minimal polynomial of b over L has coefficients in E_0 , and $N_{L(b)/L}(b) \in E_0$.*

Proof. Let $M/\mathbb{Q}$ be Galois containing L and b , $\Gamma = \text{Gal}(M/\mathbb{Q})$, $N = \text{Gal}(M/L) \trianglelefteq \Gamma$, $J = \text{Gal}(M/\mathbb{Q}(b))$. The compositum $L(b)$ corresponds to $N \cap J$ and the intersection E_0 to NJ , which is a subgroup because N is normal. Hence $[\mathbb{Q}(b) : E_0] = [NJ : J] = [N : N \cap J] = [L(b) : L] = t$ and $\deg(b) = t[E_0 : \mathbb{Q}]$. The roots of the minimal polynomial of b over L form the N -orbit of b ; an element of J fixes b and permutes this orbit (normality of N), so the coefficients of that polynomial, being symmetric functions of the orbit, are fixed by NJ and lie in E_0 . The norm is the constant coefficient up to sign. $\square$

Theorem 8.2 (two-factor criterion). *Let $L/\mathbb{Q}$ be finite Galois, $0 \neq a \in L$, $n = \deg(a)$, and $d \geq 1$. There exist algebraic b, c with $a = bc$ and $\deg b, \deg c \leq d$ if and only if there are subfields $E, F \subseteq L$ and an integer $t \geq 1$ with*

$$t[E : \mathbb{Q}] \leq d, \quad t[F : \mathbb{Q}] \leq d, \quad E \cap a^t F \neq \{0\}. \quad (8)$$

It suffices to consider $1 \leq t \leq \min(d, \lfloor d^2/n \rfloor)$, and pairs with $n \leq t[E : \mathbb{Q}][F : \mathbb{Q}]$.

Proof. Necessity. If $a = bc$ then $L(b) = L(c)$ because $c = a/b$ and $b = a/c$; call it M and put $t = [M : L]$, $E = L \cap \mathbb{Q}(b)$, $F = L \cap \mathbb{Q}(c)$. By Lemma 8.1, $\deg b = t[E : \mathbb{Q}] \leq d$ and $\deg c = t[F : \mathbb{Q}] \leq d$, and $u = N_{M/L}(b) \in E$, $v = N_{M/L}(c) \in F$ are nonzero with $uv = N_{M/L}(a) = a^t$. So $u = a^t/v \in E \cap a^t F$.

Sufficiency. Let $0 \neq u \in E \cap a^t F$, say $u = a^t w$ with $w \in F$. Choose any t th root b of u and put $c = a/b$. Then $b^t = u \in E$ and $c^t = a^t/u = 1/w \in F$, so b and c are roots of $X^t - u \in E[X]$ and $X^t - 1/w \in F[X]$ and $\deg b \leq t[E : \mathbb{Q}] \leq d$, $\deg c \leq t[F : \mathbb{Q}] \leq d$. The product is a by construction; no root-of-unity ambiguity arises because c is defined by division.

Bounds. If (8) holds then $a^t \in EF$, so $n \leq t \deg(a^t) \leq t[EF : \mathbb{Q}] \leq t[E : \mathbb{Q}][F : \mathbb{Q}] \leq d^2/t$; and $t \leq d$ since $t[E : \mathbb{Q}] \leq d$. $\square$

The sharper necessary inequality $n \leq t[EF : \mathbb{Q}]$ is useful computationally. If $E = L^H$ and $F = L^J$, then $EF = L^{H \cap J}$ and $[EF : \mathbb{Q}] = [G : H \cap J]$, so intersecting the two finite subgroups can reject a pair before any rational null-space calculation. This improves the product-of-degrees bound when the fields overlap, including all nested pairs.

Corollary 8.3. $D_{\times,2}(a)$ is computable: test $d = \max(\lceil \sqrt{n} \rceil, \text{lower bounds}), \dots, n-1$ in turn, for each d the exponents t and all pairs of subfields of L ; the first success gives $D_{\times,2}(a)$ and a decomposition, and if none succeeds then $D_{\times,2}(a) = n$. With $t = 1$ and both fields inside a fixed number field K (not necessarily Galois), the same test decides whether $a \in E^\times F^\times$, hence computes the two-factor minimum inside K . If K is Galois, testing all the permitted exponents computes $D_{\times,2}(a)$ globally; restricting to $t = 1$ still only decides the problem inside K .

The condition $E \cap a^t F \neq \{0\}$ is rational linear algebra: with column bases B_E, B_F and the matrix M_{a^t} of multiplication by a^t , it holds if and only if $[B_E \quad -M_{a^t} B_F]$ has a nonzero null vector (x, y) ; then $u = B_E x \neq 0$ (else $y = 0$ too), and $b = u^{1/t}$, $c = a/b$. No unit groups, ideal factorizations or logarithms are needed for two factors. For the product example (1) the pair $(\mathbb{Q}(u), \mathbb{Q}(v))$ with $t = 1$ succeeds at $d = 3$, which is also the lower bound; hence $D_\times(\alpha_\times) = D_{\times,2}(\alpha_\times) = 3$.

8.1 Optimal factors outside the splitting field

Theorem 8.4. Let $a = \sqrt{(1 + \sqrt{2})(1 + \sqrt{3})} > 0$, with minimal polynomial $x^8 - 4x^6 - 16x^4 - 8x^2 + 4$ (it is `Root[... , 4]`). Then $D_\times(a) = D_{\times,2}(a) = 4$, attained by $a = \sqrt{1 + \sqrt{2}} \cdot \sqrt{1 + \sqrt{3}}$ with quartic factors of minimal polynomials $x^4 - 2x^2 - 1$ and $x^4 - 2x^2 - 2$, and $D_+(a) = 8$. Moreover no finite sum or product of elements of the splitting field L of a of degree less than 8 equals a ; the optimal factors lie outside L , and at the optimal bound $d = 4$ the criterion (8) holds only with $t = 2$.

Proof. Write $r = \sqrt{2}$, $s = \sqrt{3}$, $A = 1 + r$, $B = 1 + s$, $B_0 = \mathbb{Q}(r, s)$. The four conjugates of $a^2 = AB$ are distinct, so $\mathbb{Q}(a^2) = B_0$; AB is negative at some real embedding of the totally real field B_0 , so it is not a square there and $\deg a = 8$. Put $B_1 = B_0(i)$ and $L = B_1(a)$. Since a is real and $B_1 \cap \mathbb{R} = B_0 \not\ni a$, $[L : \mathbb{Q}] = 16$. The conjugates of a are $\pm a, \pm iB/a, \pm irA/a, \pm r/a$: each squares to a conjugate of AB , for instance $(iB/a)^2 = -B^2/(AB) = -B/A = B(1 - r) = (1 + s)(1 - r)$ because $(1 + r)(1 - r) = -1$; all of them lie in L ; conversely the splitting field contains $a \cdot iB/a = iB$, hence i , and B_0 , so L is exactly the splitting field. Let z fix B_1 and send $a \mapsto -a$; let σ, τ, κ be the involutions acting by $(r, s, i, a) \mapsto (-r, s, i, iB/a)$, $(r, -s, i, irA/a)$ and $(r, s, -i, a)$ (complex conjugation). Direct substitution shows z is central, $[\sigma, \tau] = [\sigma, \kappa] = [\tau, \kappa] = z$, every element is uniquely $\sigma^e \tau^f \kappa^g z^h$ with exponents in $\{0, 1\}$, and $(\sigma^e \tau^f \kappa^g z^h)^2 = z^{ef+eg+fg}$; so $G = \text{Gal}(L/\mathbb{Q})$ has order 16 and exponent 4.

Every subgroup of order at least 4 contains z . Elements of order 4 square to z . A subgroup of order 4 avoiding z would be a Klein four-group of non-central involutions; these lie in the three cosets $\sigma\langle z \rangle, \tau\langle z \rangle, \kappa\langle z \rangle$, involutions from different cosets do not commute, and two from the same coset multiply to z . Every larger 2-group contains a subgroup of order 4.

Consequently every subfield of L of degree 1, 2 or 4, i.e. of degree less than 8, is contained in $L^{\langle z \rangle} = B_1$, and since $a \notin B_1$ no sum or product of such elements is a . Trace projection (Lemma 6.1) moves any additive decomposition into L without increasing degrees, so $D_+(a) = 8$. The displayed factorization gives $D_\times(a) \leq 4$, and Theorem 5.1(ii) with exponent $4 \nmid \text{lcm}(1, 2, 3) = 6$ gives $D_\times(a) \geq 4$. In the criterion, $E = \mathbb{Q}(\sqrt{2})$, $F = \mathbb{Q}(\sqrt{3})$, $t = 2$, $u = 1 + \sqrt{2}$ works: $a^2/u = 1 + \sqrt{3} \in F$. Every pair with $t = 1$ fails because the fields of degree at most 4 lie in B_1 . $\square$

The reports [3, 5, 9] point out that Theorem 3.1 of the manuscript [12], as they read it, asserts that two lower-degree factors can always be replaced by lower-degree factors inside the normal closure, and that its proof divides by a coefficient that can vanish (the linear coefficient of $X^2 - (1 + \sqrt{2})$ here). We have not verified the manuscript independently; Theorem 8.4 shows that any such descent statement without the exponent t is false. A related norm-and-radical replacement, with an explicit root of unity, appears in Samuels's work on the metric Mahler

measure [13]; Mahler measure ignores roots of unity, absolute degree does not, so that result does not transfer.

9 Products of arbitrarily many factors

9.1 Two factors are not enough

Example 9.1. $\eta_3 = (1 + \sqrt{2})(1 + \sqrt{3})(1 + \sqrt{5})$ has degree 8 (all eight coordinates in the multiquadratic basis are nonzero, so no sign change fixes it) and $D_\times(\eta_3) = 2$ by the prime bound. But two factors of degree at most 3 cannot work: the compositum of two fields of degree at most 3 has degree in $\{1, 2, 3, 4, 6, 9\}$, never a multiple of 8, so it cannot contain η_3 . Hence $D_{\times,2}(\eta_3) = 4$, attained e.g. by $(1 + \sqrt{2})(1 + \sqrt{3}) \cdot (1 + \sqrt{5})$.

Example 9.2. $w = (1 + \sqrt{2})(1 + \sqrt{3})(1 + \sqrt{6}) = 7 + 4\sqrt{2} + 3\sqrt{3} + 2\sqrt{6}$ has degree 4 and minimal polynomial $x^4 - 28x^3 + 128x^2 - 200x + 100$. $D_\times(w) = 2$, but $D_{\times,2}(w) = 4$: in $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ with $d = 3$, the only permitted exponents are $t = 1, 2$. The case $t = 2$ forces $E = F = \mathbb{Q}$ and $w^2 \in \mathbb{Q}$, which is false. For $t = 1$, the three pairs of quadratic subfields are excluded because the product $(x_0 + x_1\sqrt{m})(y_0 + y_1\sqrt{m'})$ has rank-one coefficient matrix, while the matrices of w are $\begin{pmatrix} 7 & 3 \\ 4 & 3/2 \end{pmatrix}$, $\begin{pmatrix} 7 & 2 \\ 4 & 3/2 \end{pmatrix}$, $\begin{pmatrix} 7 & 2 \\ 3 & 4/3 \end{pmatrix}$ with determinants $2, \frac{5}{2}, \frac{10}{3}$. Here even a greedy strategy fails: no factorization of w into two factors of smaller degree exists at all, yet three quadratic factors do.

Example 9.3. For c the real root of $x^3 + x + 3$, the product uvc of three cubic roots has degree 27 and $D_\times = 3$ (prime bound); no two factors of degree ≤ 3 can reach degree $27 > 9$ [4].

9.2 Norms, radicals and the root-of-unity obstruction

Proposition 9.4 (necessary condition). *Let $L/\mathbb{Q}$ be Galois, $0 \neq a \in L$, and $a = b_1 \cdots b_r$ with $\deg b_i \leq d$. Put $E_i = L \cap \mathbb{Q}(b_i)$, $t_i = [L(b_i) : L]$, $n_i = N_{L(b_i)/L}(b_i) \in E_i$ and $T = [L(b_1, \dots, b_r) : L]$. Then $t_i[E_i : \mathbb{Q}] \leq d$, $t_i \mid T$, and*

$$a^T = \prod_{i=1}^r n_i^{T/t_i}. \quad (9)$$

Proof. Take norms from $M = L(b_1, \dots, b_r)$ to L : $N_{M/L}(b_i) = N_{L(b_i)/L}(b_i)^{[M:L(b_i)]}$ by transitivity, and $N_{M/L}(a) = a^T$. The degree statements are Lemma 8.1. $\square$

Proposition 9.5 (sufficient condition). *If there are subfields $E_i \subseteq L$, integers $t_i \geq 1$ with $t_i[E_i : \mathbb{Q}] \leq d$, and $n_i \in E_i^\times$ with $a = \prod_i n_i^{1/t_i}$ for some choice of the radicals, then $D_\times(a) \leq d$.*

Proof. $b_i = n_i^{1/t_i}$ satisfies $X^{t_i} - n_i \in E_i[X]$, so $\deg b_i \leq t_i[E_i : \mathbb{Q}] \leq d$. $\square$

The gap between the two propositions is a root of unity. From (9), the radicals $b'_i = n_i^{1/t_i}$ satisfy $\prod b'_i = \zeta a$ with $\zeta^T = 1$; if $\tau = \text{lcm}(t_i)$ then $\zeta^\tau = \prod n_i^{\tau/t_i} / a^\tau \in L$, so ζ is a root of unity of order dividing $\tau |\mu(L)|$. The radicals may be adjusted by t_i th roots of unity, which absorb ζ when $\zeta \in \mu_\tau$, but not in general: $\zeta_8 = (1 + i)/\sqrt{2}$ is a product of numbers of degree 2 although $\deg \zeta_8 = 4$, so a stray root of unity cannot simply be appended as a low-degree factor. For two factors this issue does not arise ($t_1 = t_2 = T$, and c is defined by division), which is why Theorem 8.2 is complete.

Further problem 9.6. Construct a terminating decision procedure for $D_\times(a)$. Equivalently, given a Galois number field L , $a \in L^\times$ and d , decide whether a lies in the subgroup of $\overline{\mathbb{Q}}^\times$ generated by the multiplicative groups of all number fields of degree at most d . Even the restricted question with all factors in L and $t = 1$ — is $a \in \prod_{[E:\mathbb{Q}] \leq d} E^\times$? — is a question about multiplicative groups of number fields (units, class groups, S -units), not about vector spaces. No

complete criterion for either many-factor problem is established here, and this formulation is not a claim that their decidability is an established open problem in the literature. A positive semi-decision procedure exists by enumerating the number of factors, degrees, coefficient heights and branches and checking the resulting identities exactly; the present methods do not supply a terminating negative certificate in general.

9.3 Arithmetic obstructions beyond Galois theory

Theorem 9.7 (sign obstruction, [6]). *Let $\theta = 1 + \sqrt{2} + \sqrt{3}$, of degree 4 with minimal polynomial $x^4 - 4x^3 - 4x^2 + 16x - 8$ and norm $N_{\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}}(\theta) = -8$. Then $D_+(\theta) = 2$ and $D_\times(\theta) = 4$.*

Proof. $D_+(\theta) \leq 2$ is visible and $\theta \notin \mathbb{Q}$. For products we show that no finite product of numbers of degree at most 3 equals θ .

Step 1: cubic factors can be removed after an odd power. Let $\theta = \prod \beta_j$ with $\deg \beta_j \leq 3$. Let K be the compositum of $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, the quadratic fields of the quadratic β_j , and the quadratic resolvent fields of the cubic β_j ; K is multiquadratic, of degree a power of two. A cubic β_j has $[K(\beta_j) : K] = 3$ (a cubic with a root in a field of 2-power degree is impossible), its minimal polynomial over K is its rational cubic, and $N_{K(\beta_j)/K}(\beta_j) = N_{\mathbb{Q}(\beta_j)/\mathbb{Q}}(\beta_j) \in \mathbb{Q}^\times$. Let E be the compositum of K with the splitting fields of the cubics; each is cyclic of degree 3 over K because the resolvent is already in K , so $[E : K] = 3^s =: m$ is odd. Taking $N_{E/K}$ of the product gives $\theta^m = q \prod_{\deg \beta_j \leq 2} \beta_j^m$ with $q \in \mathbb{Q}^\times$: an odd power of θ is a product of numbers of degree at most 2.

Step 2: a product of quadratics in a totally real biquadratic field has positive norm. Let $\theta' \in L$ be a finite product of numbers of degree ≤ 2 . The non-real quadratic factors have a real product (the other factors and θ' are real), and each non-real quadratic β has $\beta\bar{\beta} \in \mathbb{Q}_{>0}$, so the square of their product is a positive rational and they may be replaced by a single real quadratic factor. Now all factors are real; let $M \supseteq L$ be a totally real multiquadratic field containing them. For a real quadratic v the function $\sigma \mapsto \text{sign}(\sigma v)$ on $\text{Gal}(M/\mathbb{Q})$ is a constant times a character with values ± 1 , and this form is preserved under products. Since $\theta' \in L$, the resulting function is constant on cosets of $\text{Gal}(M/L)$ and descends to $\text{Gal}(L/\mathbb{Q}) = C_2 \times C_2$, where a constant has 0 or 4 negative values and a nontrivial character exactly 2; so the product of the signs of the four conjugates of θ' is $+1$, i.e. $N_{L/\mathbb{Q}}(\theta') > 0$.

Applying Step 2 to $\theta' = \theta^m$ gives $N(\theta^m) = (-8)^m < 0$, a contradiction. Hence no product of numbers of degree ≤ 3 equals θ , and $D_\times(\theta) = 4$. $\square$

Corollary 9.8. *If L is a totally real biquadratic field and $a \in L$ has $N_{L/\mathbb{Q}}(a) < 0$, then $D_\times(a) = 4$.*

Proof. The same odd-power descent and sign argument excludes every product of numbers of degree at most 3. The one-factor representation gives the opposite bound, since $\deg(a) \leq [L : \mathbb{Q}] = 4$. $\square$

Here the Galois group is $C_2 \times C_2$: the prime bound and the exponent bound give 2, yet $D_\times = 4$. Any complete multiplicative algorithm must see arithmetic (here the sign of a norm), which is why linear algebra alone cannot solve the general many-factor problem.

9.4 An exact criterion for linearly disjoint families

Proposition 9.9 (rank-one tensor test, [11]). *Let K be a number field, $a \in K^\times$, and $E_1, \dots, E_r \subseteq K$ subfields with $\prod_i [E_i : \mathbb{Q}] = [K : \mathbb{Q}]$ such that the products of basis elements span K (equivalently, the multiplication map $E_1 \otimes_{\mathbb{Q}} \dots \otimes_{\mathbb{Q}} E_r \rightarrow K$ is an isomorphism). Write $a = \sum c_{j_1 \dots j_r} e_{1,j_1} \dots e_{r,j_r}$ in the product basis. Then $a = u_1 \dots u_r$ with $u_i \in E_i$ if and only if the tensor c has rank one, which is decided by the pivot test: for a nonzero entry c_p , c has rank one if and only if*

$$c_{j_1 \dots j_r} = c_p \prod_{i=1}^r \frac{c_{p(i \leftarrow j_i)}}{c_p} \quad \text{for every } (j_1, \dots, j_r),$$

where $p(i \leftarrow j)$ is p with its i th index replaced by j . The factors are then read off from the slices through p , absorbing the leading scalar c_p into one factor. For $a = 0$ the decomposition is immediate and no nonzero pivot is needed.

Proof. The multiplication map is an isomorphism of $\mathbb{Q}$ -spaces, so a is a product of elements of the E_i if and only if its tensor is a pure tensor, and the pivot formula characterizes pure tensors. $\square$

For η_3 of Example 9.1, $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ and the three quadratic subfields satisfy the hypothesis; the tensor is $(1, 1) \otimes (1, 1) \otimes (1, 1)$ and the three factors are recovered. For w of Example 9.2 the three quadratic fields have $2 \cdot 2 \cdot 2 \neq 4$, the hypothesis fails, and indeed the test does not apply; there a bounded dictionary search finds the three factors.

9.5 What the implementation does for many factors

In order of cost: the complete two-factor algorithm; the tensor test over all families of subfields of L (or of K) whose degrees multiply to the ambient degree; recursive splitting of the factors found (a heuristic: a two-factor optimum need not refine to a many-factor optimum, but it often does, as for $\eta_3 = [(1 + \sqrt{2})(1 + \sqrt{3})] \cdot (1 + \sqrt{5})$); and a small bounded dictionary search. A result is marked globally optimal only when its maximum degree equals a proved lower bound, and *two-factor optimal* when the complete two-factor search has been exhausted.

10 Comparison of the nine reports

All nine reports establish the same core: the resultant identities, irreducibility and branch selection for the examples, the largest-prime bound, the trace descent and the fixed-space criterion for sums, the intersection criterion $E \cap aF \neq 0$ for two factors in a fixed field, and the statement that arbitrary-length products are not covered by a complete algorithm. Table 1 records what each report contributed beyond that. Every report states that its Wolfram code was not executed (the evaluator they had access to returned HTTP 404); the Python/SymPy checks they ran verify the algebraic identities, not the packages.

Table 1: What the nine reports added to the common core.

Report	Distinctive contributions and remarks
[3]	Conjugate-subset criterion (Theorem 6.3) with the equivariant-projection proof; a degree-eight counterexample $\sqrt{1 + \sqrt{2}}\sqrt{1 + \sqrt{7}}$ with group of order 32 (an extraspecial central product) and the same conclusions as Theorem 8.4; template (Gröbner) recovery of the cubics.
[4]	Residual-resultant search with a height bound on one polynomial only; the complete two-factor criterion with exponent k ; the degree-27 example of three cubics; the $2^{1/4}$ example ($D_+ = 4$).
[5]	Two-factor criterion with exponent t and the pruning bound $n \leq t[E : \mathbb{Q}][F : \mathbb{Q}]$; the example $w = (1 + \sqrt{2})(1 + \sqrt{3})(1 + \sqrt{6})$ with $D_{\times,2} = 4 > D_{\times} = 2$ and the determinant proof; a vanishing-linear-coefficient example $\sqrt{\sqrt{2}(1 + \sqrt{3})} = 2^{1/4}\sqrt{1 + \sqrt{3}}$; subfields by coefficient fields of divisors.
[6]	Inverse composition with a fixed first polynomial; the sign obstruction (Theorem 9.7); the $x^4 - x - 1$ pair-sum example (Example 6.6); the ζ_5 example; fixed fields by iterated intersection with $\ker(A_{\sigma} - I)$.
[7]	Principal subfields $V_g = \{h(\theta) : g \mid h(x) - h(\theta)\}$ with a self-contained proof (after [16]); rational normalization of the recovered pair; explicit recovery formulas.
[8]	Subfields by coefficient fields of divisors; trace centering; the sextic additive optimum $D_+(uv) = 6$ with the projection $T = (1 - e_1)(1 - e_2)$ argument; the multiplicative subgroup formulation $B_d(K)$.

Report	Distinctive contributions and remarks (continued)
[9]	The degree-eight example of Theorem 8.4 with a complete group-theoretic proof; the exponent bound with $\text{lcm}(1, \dots, d)$; the three-quadratic example η_3 ; the many-factor norm relation (9) and the ζ_8 remark; the most complete package design (never executed).
[10]	The nonmembership certificate (a separating functional $w^\top B_d = 0$, $w^\top a \neq 0$); the $\sqrt{2 + \sqrt{2}}$ example; the necessary norm-descent condition (saturation of $B_d(L)$).
[11]	Rank-one tensor test (Proposition 9.9); polynomial recovery certificates $A_\times, B_\times, A_+$; the sextic pair-product example $r_1 r_2$ with $D_\times^{\mathbb{Q}(r_1 r_2)} = 6 > 4 = D_\times$; the example $\sqrt{2} + \sqrt{3} + \sqrt{6}$ as a three-term sum not reducible to two terms.

Points on which the reports are inconsistent with one another, or which the present work corrects:

- The reports describe the degree-eight example either as a counterexample to a published descent theorem [3, 9] or only as a gap in its proof [5]. The mathematical content (Theorem 8.4) is not in dispute; we have phrased the attribution cautiously.
- All reports construct the splitting field with `ToNumberField[roots, All]` and convert elements with `ToNumberField[a, theta]`. On the examples this takes 14 minutes for the first call and 10 minutes per element conversion in Wolfram 15.0.1 (Section 11); it is not a usable design.
- Several reports enumerate subfields of a Galois field by subsets of the linear factors of the minimal polynomial ($2^{[L:\mathbb{Q}]-1}$ subsets) or by all subgroups without first computing the group; the numerical-resolvent approach below is far cheaper.
- Report [11] notes that the binary product search of its package is complete only inside $\mathbb{Q}(a)$ and that $\sqrt{2} + \sqrt{3} + \sqrt{6}$ needs three terms; Section 7 strengthens this to a theorem covering both operations and Gaussian coefficients.

11 Algorithms and implementation

11.1 Why not the power basis of a primitive element

All reports propose: compute all roots, call `ToNumberField[roots, All]` to obtain a primitive element θ of L with the roots as polynomials in θ , factor the minimal polynomial of θ over L to get the automorphisms, then enumerate subgroups. For the degree-nine examples ($[L:\mathbb{Q}] = 36$) the first call took about 14 minutes in Wolfram 15.0.1 and a single `ToNumberField[root, theta]` conversion a further 10 minutes. The obstacle is not the degree but the power basis: a generic primitive element $\theta = \sum w_i \alpha_i$ has a minimal polynomial with coefficients around 10^{20} , and the coordinates of the roots in the basis $1, \theta, \dots, \theta^{35}$ have enormous height. By contrast, `RootReduce[theta + w*Root[p,k]]`, which adjoins one root to a partial primitive element, takes about a second per step. The implementation therefore never uses the power basis of θ .

11.2 The Galois group by numerical resolvents

Let P be monic with integer coefficients (replace a by ca , c the leading coefficient of its minimal polynomial, so that all roots are algebraic integers) and let $\beta_1, \dots, \beta_n$ be its roots, computed to high precision. Choose integer weights $w_1, w_2, \dots$ and consider the partial sums $\theta_k = \sum_{i \leq k} w_i \beta_i$. The Galois group $G \leq S_n$ is determined level by level: the set O_k of k -tuples τ with $(\beta_{\tau(1)}, \dots, \beta_{\tau(k)})$ in the G -orbit of $(\beta_1, \dots, \beta_k)$ is exactly the set of τ for which $\theta_\tau = \sum_{i \leq k} w_i \beta_{\tau(i)}$ is a root of the minimal polynomial m_k of θ_k , provided the map $\tau \mapsto \theta_\tau$ is injective on the candidate set $\{(\tau', j) : \tau' \in O_{k-1}, j \notin \tau'\}$. The candidate set is G -stable, so

$F_k(x) = \prod_{\text{candidates}} (x - \theta_\tau)$ is a monic integer polynomial; it is squarefree exactly when the injectivity holds, and then m_k is its irreducible factor vanishing at θ_k . At level n , $O_n = G$ as a set of permutations, and closure under composition is verified.

The Python implementation does this rigorously in Arb ball arithmetic [20]: F_k is computed as a ball polynomial, each coefficient ball must contain a unique integer (otherwise the precision is doubled), F_k is factored exactly over $\mathbb{Z}$ with FLINT [21], and $\tau \in O_k$ is tested by $m_k(\theta_\tau) \ni 0$; since every true orbit element passes this test and the number of passing candidates is checked to equal $\deg m_k$, the orbit is determined exactly. The Wolfram implementation obtains m_k from `RootReduce` of $\theta_{k-1} + w_k \text{Root}[P, k]$ (exact), locates the roots numerically and applies the same counting checks; a degenerate weight (degree drop or collision) is detected and redrawn. The dominant cost is the exact minimal polynomial (or the exact factorization of F_k , of degree at most $n|G|$); the method is practical for $|G|$ up to a few hundred.

11.3 The splitting field in a tower basis with trace coordinates

The levels at which $\deg m_k$ grows give a tower $L = \mathbb{Q}(\beta_{g_1}, \dots, \beta_{g_s})$ with relative degrees $e_1, \dots, e_s$, $\prod e_j = |G| = N$, and the monomials $b = \prod_j \beta_{g_j}^{a_j}$, $0 \leq a_j < e_j$, form a $\mathbb{Q}$ -basis of L consisting of algebraic integers of small height. Every $\sigma \in G$ acts on such a monomial by permuting the roots, so the values $\sigma(b)$ are known numerically for all σ and all basis elements: an $N \times N$ matrix `Val`. The trace form $\text{Tr}_{L/\mathbb{Q}}(b_j b_l) = \sum_\sigma \sigma(b_j) \sigma(b_l)$ is an integer matrix $\text{Gram} = \text{Val}^\top \text{Val}$ (with a transpose, not a conjugate transpose), nondegenerate since $L/\mathbb{Q}$ is separable, and the coordinates of any $y \in L$ given by its conjugate vector $(\sigma(y))_\sigma$ are $\text{Gram}^{-1}(\text{Tr}(y b_l))_l$. When y is integral these traces are integers; for a rational combination of the integral basis elements, first clear its coordinate denominators, recover integer traces, then divide back. A unique integer in each Arb ball certifies the recovery in Python; ordinary high-precision rounding in Wolfram has the qualification described below. This yields, by rational linear algebra with integer inputs: the coordinates of all roots, the matrix A_σ of every automorphism (conjugate vector of $\sigma(b_j)$ is $\tau \mapsto (\tau\sigma)(b_j)$, read from the multiplication table of G), multiplication matrices, fixed fields $L^H = \bigcap_{h \in H} \ker(A_h - I)$ for all subgroups (enumerated by closure from cyclic subgroups), and the linear systems of Theorems 6.2 and 8.2. An element with coordinate vector v is converted back to a `Root` object by computing its distinct conjugates (the orbit of v under the A_σ), forming their polynomial after clearing coordinate denominators, and identifying the intended branch. In Python the branch must be the unique isolated root compatible with the element's enclosure; precision is increased when the match is ambiguous. A consistency check (A_σ permutes the root coordinates as σ does) guards the numerical steps in Wolfram. Its final identities and degrees are checked by `RootReduce` and `MinimalPolynomial`; Python also supplies an independent exact verifier using composed polynomials and certified branch isolation.

11.4 The fast path inside $\mathbb{Q}(a)$

Independently of the Galois group, all subfields of $K = \mathbb{Q}(a)$ are obtained from the factorization $P = \prod g_i$ of the minimal polynomial over K (`Factor[P, Extension -> Root[P, k]]`, 0.3 s for the examples): each factor g gives the principal subfield $V_g = \{h(\theta) : g \mid h(x) - h(\theta)\}$, computed as a null space, and every subfield is an intersection of principal subfields [16, 7, 11].

For completeness, the field property and the intersection assertion have a short explanation. A polynomial $g \mid P$ imposes $h(\beta) = h(\theta)$ for its roots β , with h taken modulo P ; these equalities are preserved by addition, multiplication and inversion of a nonzero element, so V_g is a subfield of K . If $E \subseteq K$, the roots of the minimal polynomial of θ over E are exactly the images of θ under embeddings of K fixing E . An element $h(\theta)$ has the same value under all those embeddings if and only if it belongs to E , by separability. Factoring that minimal polynomial over K therefore expresses E as the intersection of the corresponding V_{g_i} . Closing the principal-field list under intersections therefore gives all subfields, with an exact linear-algebra construction.

In K the additive criterion is complete for sums inside K , and the two-factor criterion with $t = 1$ is complete for two factors inside K . If K is Galois (all g_i linear), the additive search is globally complete and the product search becomes globally complete for two factors by allowing all the exponents of Theorem 8.2. The option restricting components to K must still enforce $t = 1$, even for Galois K . The implementations try this path first and stop when it reaches a lower bound, which is what happens for both examples in (1): the answers (2) are found and certified in about a second, without constructing the splitting field.

11.5 Normalization of the output

Linear algebra returns arbitrary elements of the solution spaces. Summands are trace-centred (the rational mean of the conjugates, $\text{Tr}(y)/[L : \mathbb{Q}]$, is subtracted and collected into a rational offset); this recovers the trace-zero cubics of (2). If a term-count bound is requested, the offset is absorbed into one summand instead of adding a term; a rational translation does not change the field generated by that summand. For products the null vector is taken from an LLL-reduced basis of the integer null space, the factor u is rescaled by the p -adically minimal rational q making qu an algebraic integer (for each prime, the least exponent e with $v_p(c_i) + (d - i)e \geq 0$ for all coefficients c_i of the monic minimal polynomial), followed by a small search for the scaling of least coefficient height; the cofactor is divided accordingly. This turns $\frac{1}{4} \cdot \text{Root}[8 + 4\# + \#^3\&, 1] \cdot \text{Root}[-2 + \dots]$ into the factors of (2).

11.6 Interface

```
Get["RootDecomposition.wl"];
RootProductDecomposition[a]      RootProductDecomposition[a, d]
RootSumDecomposition[a]          RootSumDecomposition[a, d]
RootGaloisData[a]                RootDecompositionLowerBound[a]
RootBoundedDecomposition[a, Plus|Times, d, height, factors]
RootDecompositionVerify[a, terms, Plus|Times]
```

An association reports the components in "Terms", their "Degrees" and "MaximumDegree", and a "LowerBound". The certificate fields are "Optimal" (global optimality, with the qualifications below), "ScopeOptimal" (optimal within the requested class) and "TwoFactorOptimal". Further fields record the "NormExponent", "Verified" status, "Method" and an inactive "Expression".

To keep components inside $\mathbb{Q}(a)$, set "Scope" to "InputField". Set "Engine" to "SplittingField" to force the Galois route. Setting "Coefficients" to "GaussianRationals" enables unrestricted Gaussian sums, while "MaxTerms" restricts the number of ordinary summands. Combining Gaussian coefficients with a finite term cap is rejected because the unrestricted linear-space calculation does not solve that problem. Likewise, an explicit degree cap must either be met or produce a failure, including when a trivial representation is returned. "MaxFactors" $\rightarrow 2$ restricts to the complete two-factor algorithm; "MaxGroupOrder" and "WorkingPrecision" bound the Galois computation.

Python's two search functions are `sum_decomposition` and `product_decomposition`, both in `rootdecomp.py`. The module also exposes `galois_data` and `lower_bound`, the verifier `verify_exact`, and the input parser `parse_wolfram_root`. The ordinary decomposition scopes have the same meaning; Gaussian coefficients are a Wolfram-only feature, and the root-index portability qualification in the conventions applies to non-real branches.

11.7 Measurements

Table 2 records the original development measurements (Wolfram 15.0.1, Python 3.14 with python-flint 0.8). These historical timings are not fresh measurements of the revised code. In that run the Galois route of the Python port was faster than the Wolfram kernel because exact

factorization of the resolvents in FLINT is cheaper than `RootReduce` of the partial primitive elements, and its rational linear algebra runs in FLINT rather than in the interpreter. The fast path in $\mathbb{Q}(a)$ is the best choice for both examples.

Task	Wolfram	Python
Product example, fast path in $\mathbb{Q}(a)$, certified optimal	0.2 s	1.9 s
Sum example, fast path in $\mathbb{Q}(a)$, certified optimal	1.3 s	0.4 s
Splitting-field data of the degree-9 examples ($ G = 36$)	18–21 s	3–5 s
$D_+(\alpha_\times) = 6$ with the splitting-field data cached	0.2 s	0.15 s
$\sqrt{(1 + \sqrt{2})(1 + \sqrt{3})}$, $ G = 16$, two-factor with $t = 2$	4 s	0.4–2 s
$x^4 - x - 1$ pair sum, $ G = 24$	3 s	0.5 s
Product search for α_+ (two-factor, tensor families, recursion)	30 s	110 s
<code>ToNumberField[roots, All]</code> for the degree-9 examples	14 min	–

Table 2: Historical wall-clock times (Wolfram 15.0.1; Python 3.14 with `python-flint` 0.8 and `SymPy` 1.14). Current validation and benchmark results are recorded separately in the project documentation.

11.8 Limits and scope of the certificates

The implementations of the complete field-based searches need certified field data; the resolvent method is practical for groups of order up to a few hundred (the resolvent at level k has degree up to $n|G|$). For larger groups the fast path still returns decompositions inside $\mathbb{Q}(a)$ and the bounded search still finds small-height factors, but an unsuccessful bounded search gives no global negative statement.

Three distinct assertions need separate evidence: an identity holds for the selected branches; the reported component degrees are correct; and no better representation exists in the stated scope. The Python Galois construction uses ball arithmetic and exact factorization, and rational linear algebra is then exact. Its independent `verify_exact` reconstructs the sum or product by exact polynomial arithmetic with certified root selection; `verify_numeric` is only an enclosure compatibility diagnostic. Polynomial divisibility without branch selection, or overlapping numerical balls alone, would not prove a particular identity.

The Wolfram splitting-field route relies on exact minimal polynomials, numerical root matching with generous margins, and group-closure and automorphism-consistency checks. These are useful checks but do not replace a proof of every numerical reconstruction step. An exactly verified identity attaining the independent degree or Frobenius lower bound certifies global optimality without that reconstruction. An optimality flag relying on exhaustive Wolfram splitting-field enumeration instead is conditional on the reconstructed group and field data. Exact verification of the returned identity alone does not remove this qualification. A returned `"Optimal" -> False` means *not established*, not *suboptimal*; `"TwoFactorOptimal"` concerns the two-factor problem, and a request limiting the number of terms concerns that restricted class rather than the unrestricted minimum.

12 Conclusion

The additive problem is completely solved by a finite algorithm, and the two-factor multiplicative problem likewise; the key steps are trace descent to the splitting field and norm descent with an explicit radical exponent, both of which reduce to rational linear algebra on fixed fields. The arbitrary-length multiplicative problem is different in kind: its answer depends on arithmetic invisible to the Galois group, and the present algorithms do not decide it in general. For the two numbers of the question the answer is unambiguous — the cubic decompositions (2) are globally

optimal — and the same machinery shows that the product root needs sextic summands, that optimal factors can live outside the splitting field, and that flat sums of three quadratic numbers can be inaccessible to any binary splitting, even with Gaussian coefficients.

References

- [1] V. Reshetnikov, *Factor a polynomial Root into Roots of smallest possible degree*, Mathematica Stack Exchange, question 105933 (2016), with an answer by yarchik (2019). <https://mathematica.stackexchange.com/q/105933/7288>.
- [2] Wolfram Research, *Root*, Wolfram Language documentation, details on root indexing and the `ExactRootIsolation` option. <https://reference.wolfram.com/language/ref/Root.html>.
- [3] Report 1, *Decomposing Algebraic Numbers into Low-Degree Roots* (2026), `reports/report-01` in the repository.
- [4] Report 2, *Decomposing Algebraic Numbers into Low-Degree Roots: Certified Mathematica searches, global degree bounds, and exact normal-field algorithms* (2026), `reports/report-02`.
- [5] Report 3, *Decomposing Polynomial Roots into Low-Degree Sums and Products* (2026), `reports/report-03`.
- [6] Report 4, *Decomposing Algebraic Numbers into Low-Degree Roots: Exact Mathematica methods, global minimality certificates, and the distinction between sums and products* (2026), `reports/report-04`.
- [7] Report 5, *Decomposing algebraic numbers into low-degree Roots* (2026), `reports/report-05`.
- [8] Report 6, *Decomposing Algebraic Roots into Low-Degree Sums and Products* (2026), `reports/report-06`.
- [9] Report 7, *Decomposing Algebraic Roots into Low-Degree Sums and Products: Mathematica algorithms, optimality certificates, and the role of the splitting field* (2026), `reports/report-07`.
- [10] Report 8, *Decomposing Algebraic Roots Into Sums and Products of Minimum Degree* (2026), `reports/report-08`.
- [11] Report 9, *Decomposing Algebraic Numbers into Low-Degree Sums and Products: Exact algorithms, global degree bounds, and a Wolfram Language reference implementation* (2026), `reports/report-09`.
- [12] C. Altamirano, *A Problem in Algebraic Number Theory*, MIT UROP+ final paper (2017), as cited in [3, 5, 9].
- [13] C. L. Samuels, *The infimum in the metric Mahler measure*, arXiv:1408.4165 (2014).
- [14] A. Bostan, P. Flajolet, B. Salvy, É. Schost, *Fast computation of special resultants*, J. Symbolic Comput. 41 (2006), 1–29.
- [15] H. Cohen, *A Course in Computational Algebraic Number Theory*, Graduate Texts in Mathematics 138, Springer, 1993.

- [16] J. Szutkoski, M. van Hoeij, *The complexity of computing all subfields of an algebraic number field*, J. Symbolic Comput. 93 (2019), 161–182; arXiv:1606.01140.
- [17] M. van Hoeij, J. Klüners, A. Novocin, *Generating subfields*, J. Symbolic Comput. 52 (2013), 17–34.
- [18] R. P. Stauduhar, *The determination of Galois groups*, Math. Comp. 27 (1973), 981–996.
- [19] C. Fieker, J. Klüners, *Computation of Galois groups of rational polynomials*, LMS J. Comput. Math. 17 (2014), 141–158.
- [20] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. 66 (2017), 1281–1292.
- [21] W. Hart, *Fast Library for Number Theory: an introduction*, in: Mathematical Software – ICMS 2010, Lecture Notes in Computer Science 6327, Springer, 2010, 88–91; python-flint, <https://github.com/flintlib/python-flint>.
- [22] J. S. Milne, *Fields and Galois Theory*, course notes, version 5.10 (2022), <https://www.jmilne.org/math/CourseNotes/FT.pdf>.